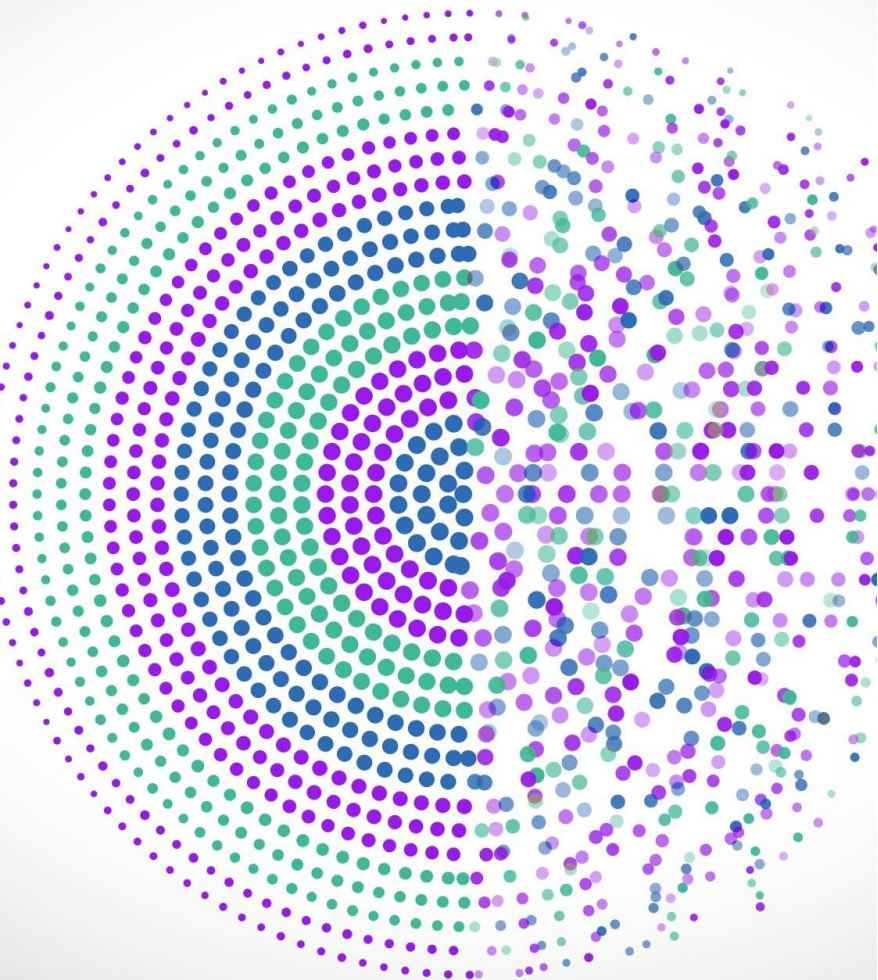




Youth Substance Use Prediction using Tree-Based Models

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DATA 5322 – Statistical Machine Learning II

Seattle University

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Objective



PREDICT YOUTH
SUBSTANCE USE



APPLY TREE-
BASED MODELS



COMPARE MODEL
PERFORMANCE



IDENTIFY KEY
PREDICTORS

Dataset and Cleaning

Replaced invalid codes with NA

Removed missing values

Prepared dataset for modeling

youth_data:

- Demographics
- Behavioral factors
- Peer influence
- Substance use



Models Used

- Decision Tree
- Random Forest
- Boosting

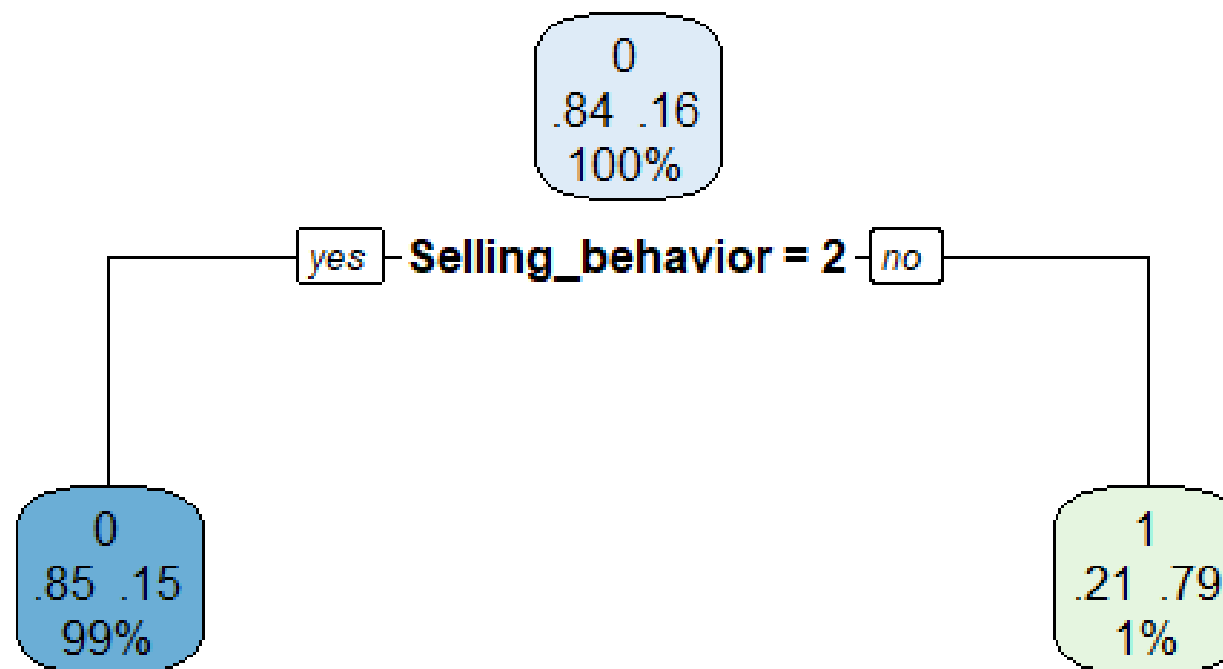
Problem Types:

- Binary classification
 - Multi-class classification
 - Regression
-

Binary Classification

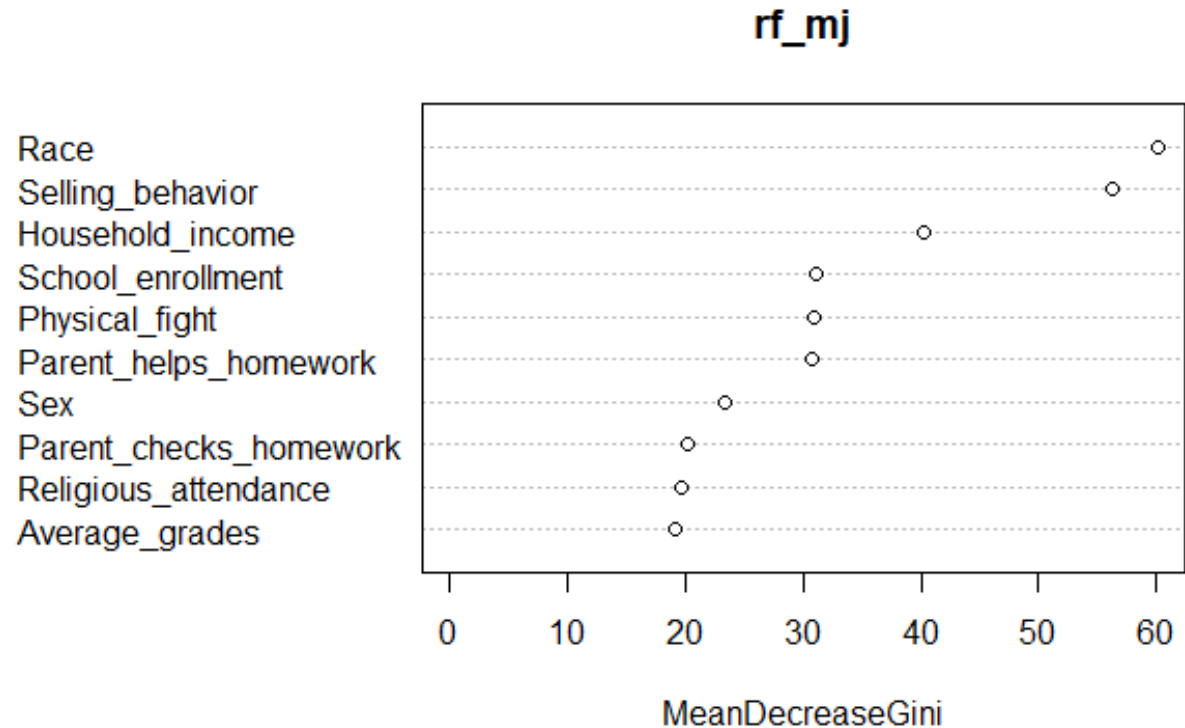
– Decision Tree

- Target: Marijuana use
- Model: Decision Tree
- Accuracy: ~85.9%
- Key predictor: Selling behavior



Random Forest (Binary)

- Target: Marijuana use
- Model: Random Forest
- Accuracy: ~85.7%
- More stable than single tree



Boosting (Binary)

Target: Marijuana use

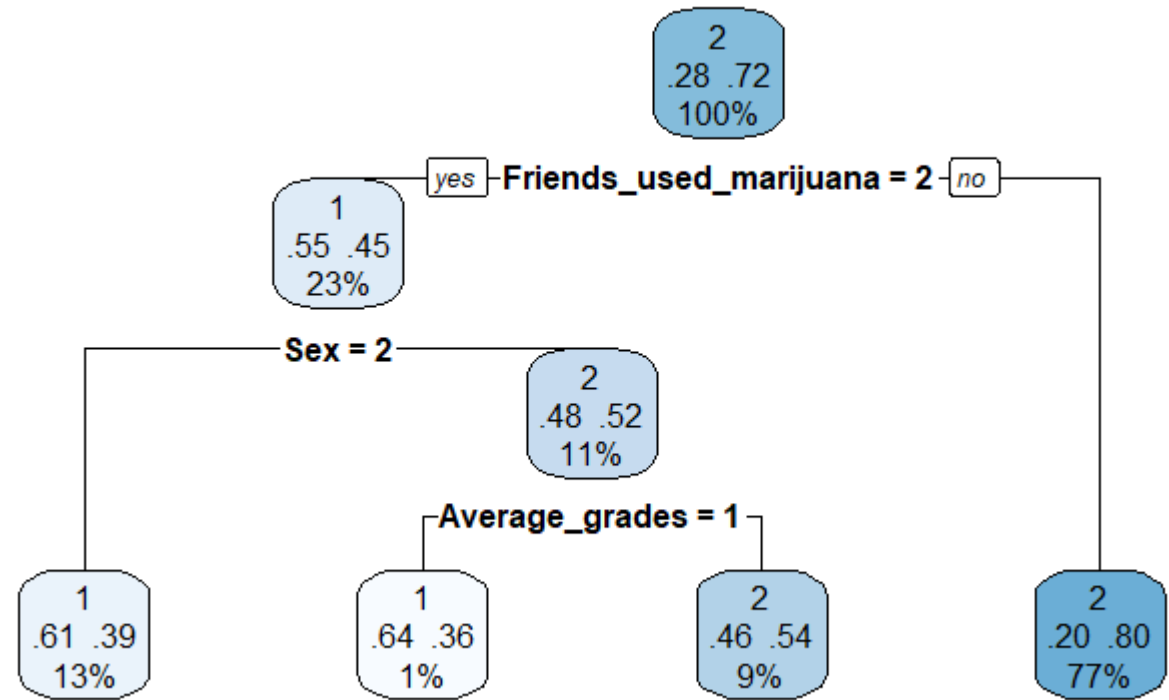
Model: Boosting

Accuracy: ~85.9%

Focuses on correcting errors

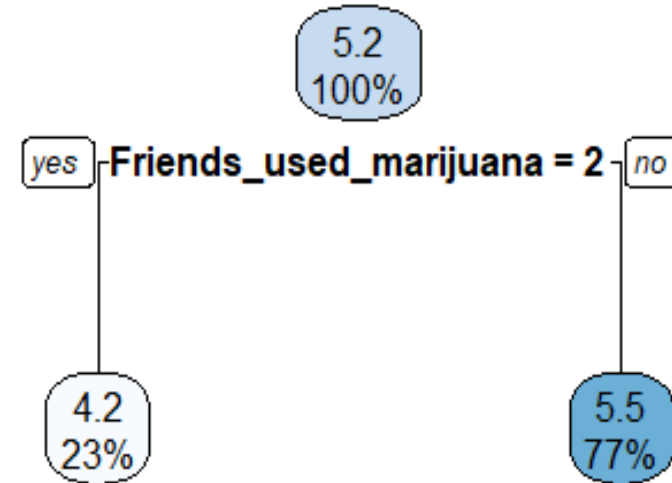
Multi-class Classification

- Target: Marijuana use level
- Model: Decision Tree
- Accuracy: ~76%
- More complex than binary



Regression Model

- Target: Alcohol use days
- Model: Regression Tree
- Metric: MSE $\approx$ 2.70
- Key predictor: Friends used marijuana

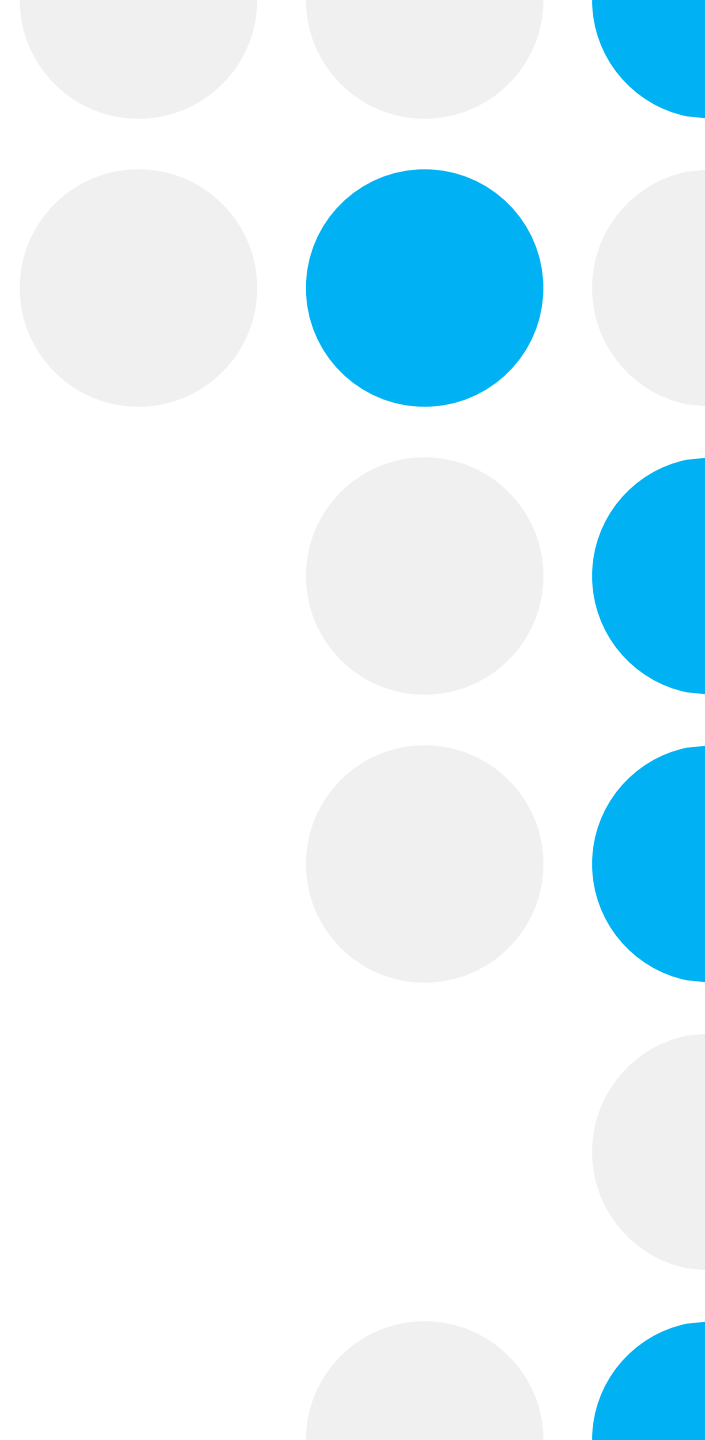


Model Comparison

Model	Value
Decision Tree	~85.9%
Random Forest	~85.7%
Boosting	~85.9%
Multi-class Tree	~76%
Regression Tree	≈ 2.70

Discussion

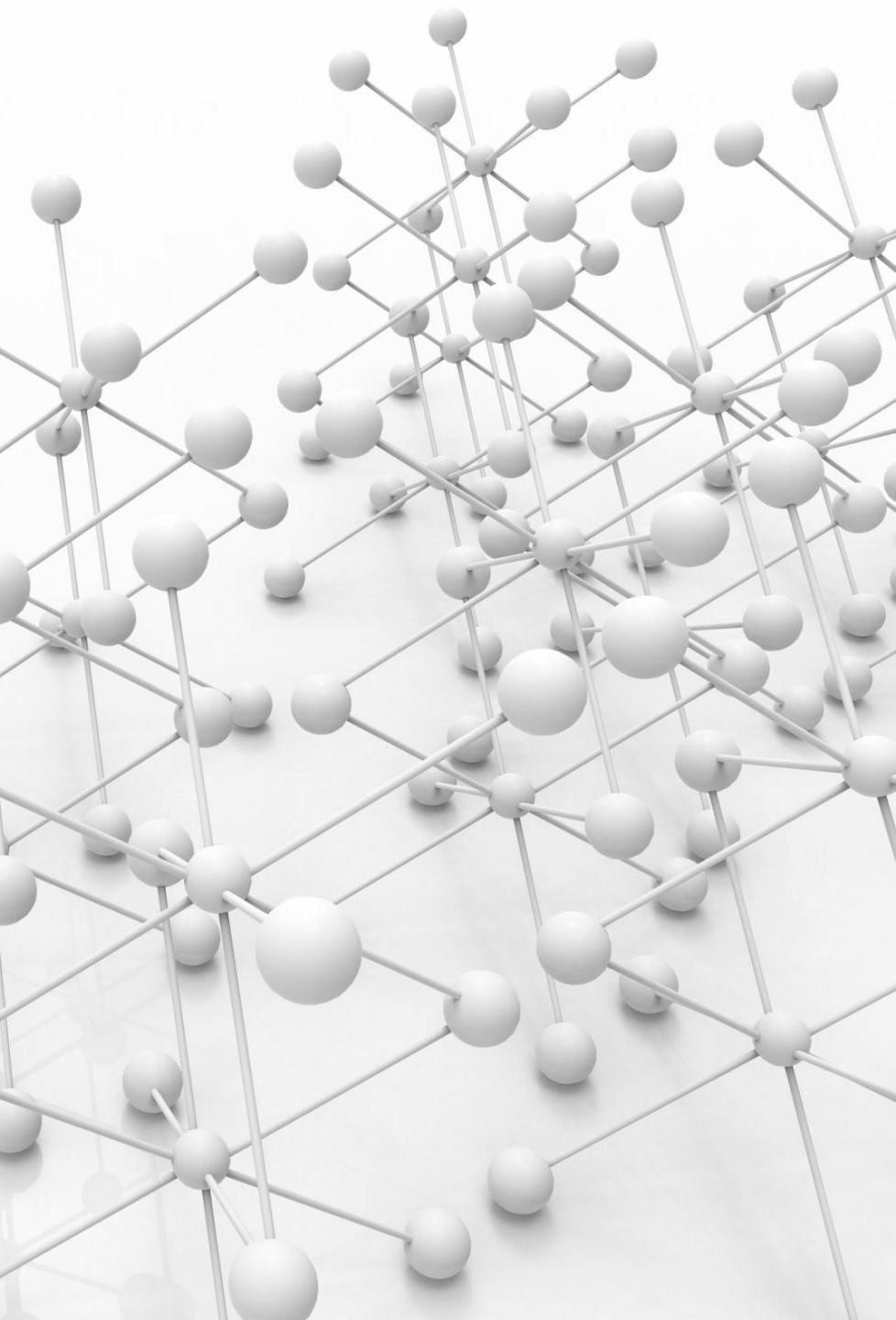
- Behavioral and peer factors are important predictors
 - Social environment plays a key role
 - Binary models perform better than multi-class
 - Multi-class prediction is more complex
 - Regression shows trends but with more variability
 - Focus on patterns, not individuals
-





Ethical Considerations

- Risk of misuse
 - Privacy concerns
 - Potential bias
 - Use for insight, not decisions
-



Conclusion

- Tree-based models performed well
- Key predictors identified
- Binary easier than multi-class
- Insights for future research



References

- Mendible, A. (n.d.).
youth_data.Rdata [Data set].
GitHub.
https://github.com/mendible/5322/blob/main/Homework%201/youth_data.Rdata
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THANK YOU!

